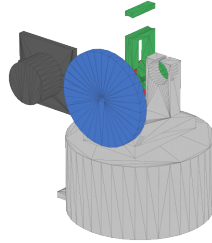


OPTION D

A Beginner's Guide to the ~1/3-Scale, 28BYJ-48 Pan/Tilt Redesign



This guide explains, in plain language, the Option D pan/tilt mechanism -- a smaller rebuild of the original **ASSEMBLY_pan_tilt** reference design, sized around a real, off-the-shelf 28BYJ-48 stepper motor (one for pan, one for tilt) instead of the original's generic placeholder motor. It has the same number of moving parts and the same two-axis mechanism as the original: it swivels left-right (**pan**) and has a second, separate motor and gear set that produces a tipping motion (**tilt**) for whatever is mounted on it. No prior mechanical design or CAD experience is assumed -- every technical word is explained the first time it is used.

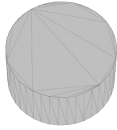
Part types / total pieces	16 types / 22 pieces
Overall assembled size (measured)	~ 109 x 137 x 109 mm (W x H x D)
Size vs. the original (245 x 298 x 270mm)	roughly 1/3 on two axes; taller relatively, driven by the real motor's fixed size (see Chapter 3)
Pan / tilt gear reduction	3 : 1 (each axis, both verified)
Motor	Real 28BYJ-48 stepper (x2) -- the original used an unbranded placeholder can

How this guide is organised: Chapter 1 introduces every physical part with a picture and a number. Chapters 2 and 3 refer back to those same part numbers throughout, so you can always flip back to Chapter 1 if you forget what "Part 6" is.

Chapter 1 - Parts of the Design

Every physical piece of this mechanism is listed below with a number, a plain-language description of what it does, and how it's meant to be made. Parts labelled **3D PRINTED** are meant to be printed on an ordinary home FDM printer; parts labelled **BOUGHT** are real, off-the-shelf items you would order rather than make. This mix (rather than the original design's machined metal) is a direct requirement of this redesign: every part must be either easily printable or purchasable, no exceptions.

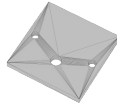
#	Part name	Type	Qty
1	Main Housing	3D PRINTED	1
2	Pan Motor Mounting Plate	3D PRINTED	1
3	Pan Platform / Tilt Tower Base	3D PRINTED	1
4	Tilt Motor Mounting Plate	3D PRINTED	1
5	Tilt Hub	3D PRINTED	1
6	Payload Fork ("Cam Holder")	3D PRINTED	1
7	Pan Gear, 60 tooth	3D PRINTED	1
8	Tilt Gear, 60 tooth	3D PRINTED	1
9	Pinion Gear, 20 tooth	3D PRINTED	2
10	28BYJ-48 Stepper Motor	BOUGHT	2
11	608ZZ Ball Bearing (pan shaft)	BOUGHT	2
12	625ZZ Ball Bearing (tilt shaft)	BOUGHT	2
13	Pan Shaft	BOUGHT	1
14	Tilt Shaft	BOUGHT	1
15	Pan Shaft Spacer	3D PRINTED	2
16	Tilt Shaft Spacer	3D PRINTED	2



1. Main Housing

3D PRINTED Qty: 1

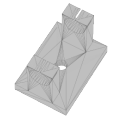
The outer shell the whole mechanism is built inside. A round-profile printed housing that carries the pan gear, both pan bearings, and the pan motor. In this layout-study model it is shown as a solid block (not hollowed out) -- good enough to check overall size and clearances, not a print-ready wall-thickness design.



2. Pan Motor Mounting Plate

3D PRINTED Qty: 1

A flat bracket that holds the pan-axis motor in a fixed position. Its screw holes and shaft clearance hole are positioned around the motor's real off-center output shaft, not its own centreline, so the shaft lines up correctly with the gear underneath.



3. Pan Platform / Tilt Tower Base

3D PRINTED Qty: 1

Sits on top of the pan shaft and rotates with it -- this is the part that carries everything else (motor, gears, shaft) around whenever the mechanism pans. Two towers on top hold the tilt shaft's two bearings.



4. Tilt Motor Mounting Plate

3D PRINTED Qty: 1

A flat bracket, mounted behind the tilt axis, that holds the tilt-axis motor. Like the pan motor's bracket, its shaft clearance hole is positioned to match that motor's own real off-center shaft, not an idealised centred one.



5. Tilt Hub

3D PRINTED Qty: 1

A small collar that clamps onto the tilt shaft with a flat-sided key and a grub screw. Carries two bolt-on flanges, one at each end, where the Payload Fork attaches.



6. Payload Fork ("Cam Holder")

3D PRINTED Qty: 1

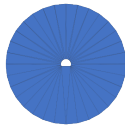
The fork-shaped bracket that carries the payload. Bolted rigidly to the Tilt Hub's two flanges -- by design decision (see Chapter 3's engineering note) it does NOT have a free-spinning joint of its own; it only moves because the Tilt Hub underneath it moves.



7. Pan Gear, 60 tooth

3D PRINTED Qty: 1

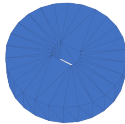
The big gear that produces the pan motion, keyed to the Pan Shaft and meshing with the pan motor's small pinion gear.



8. Tilt Gear, 60 tooth

3D PRINTED Qty: 1

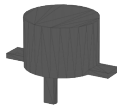
The big gear that produces the tilt motion, keyed to the Tilt Shaft and meshing with the tilt motor's small pinion gear.



9. Pinion Gear, 20 tooth

3D PRINTED Qty: 2

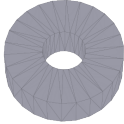
A small gear pressed onto each motor's own output shaft. One drives the pan gear, the other drives the tilt gear -- in both cases the small gear turns 3 times for every 1 turn of the big gear it drives, trading speed for turning force (see Chapter 3).



10. 28BYJ-48 Stepper Motor

BOUGHT Qty: 2

A real, off-the-shelf 28BYJ-48 stepper motor (the one named in the requirements). One drives pan, the other drives tilt. Its output shaft is not in the middle of the round motor body -- both mounting brackets (Parts 2 and 4) are shaped to line up with the real, off-center shaft position rather than an idealised centred one.



11. 608ZZ Ball Bearing (pan shaft)

BOUGHT Qty: 2

A real, off-the-shelf 608ZZ ball bearing (8mm bore) -- the same size used in skateboard wheels, cheap and easy to source. Two of these support the Pan Shaft, spaced apart for stability rather than using one large flanged bearing like the original design's.



12. 625ZZ Ball Bearing (tilt shaft)

BOUGHT Qty: 2

A real, off-the-shelf 625ZZ ball bearing (5mm bore), one pressed into each of the platform's two towers, supporting the Tilt Shaft.



13. Pan Shaft

BOUGHT Qty: 1

A short length of 8mm steel rod, cut to length and given a flat-sided section where the pan gear keys onto it. Steel rod is used here instead of a printed shaft for stiffness and wear resistance where it rides in the bearings.



14. Tilt Shaft

BOUGHT Qty: 1

A short length of 5mm steel rod, cut to length, with flat-sided sections where the tilt gear and the Tilt Hub key onto it.



15. Pan Shaft Spacer

3D PRINTED Qty: 2

A short printed sleeve that sits on the pan shaft between the gear and each bearing, keeping everything positioned correctly along the shaft's length.



16. Tilt Shaft Spacer

3D PRINTED Qty: 2

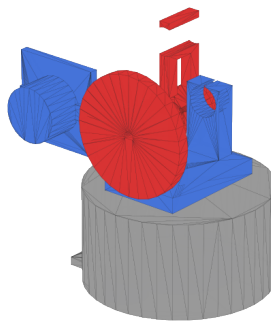
A short printed sleeve on the tilt shaft, between each bearing and the Tilt Hub, keeping everything positioned correctly along the shaft's length.

Chapter 2 - Movement

This mechanism moves in two intended directions, called axes:

Pan - swivels the top of the mechanism left and right, like shaking your head 'no'. The Pan Shaft (Part 13) and Pan Gear (Part 7) spin about a vertical axis running up through the housing. Everything mounted above them -- the whole tilt stage, Parts 3-12 and 14-16 -- is carried around with this rotation.

Tilt - a second, separate motor and gear pair tips the payload up and down, like nodding your head 'yes'. Because this tilt stage sits on top of the pan stage, it also swings left-right whenever pan happens.



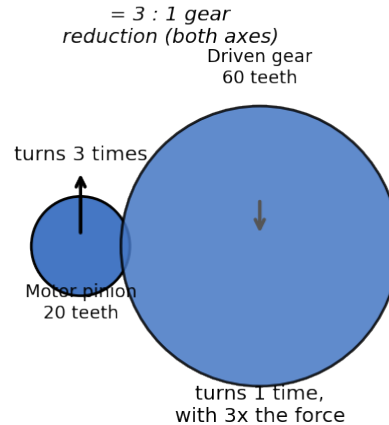
Grey = never moves (fixed) Blue = moves with PAN Red = moves with PAN + TILT

#	Part	How it moves
1	Main Housing	Never moves (fixed) -- everything else is built on or inside it.
2	Pan Motor Mounting Plate	Bolted fixed to the housing; never moves.
3	Pan Platform / Tilt Tower Base	Carried around by pan -- spinning it about the pan axis IS the pan motion.
4	Tilt Motor Mounting Plate	Carried around by pan (bolted to the platform); does not itself tilt.
5	Tilt Hub	Carried around by pan; pivots with tilt (it IS the tilt motion, along with the tilt shaft/gear).
6	Payload Fork ("Cam Holder")	Carried around by pan and tilt; rigidly bolted to the Tilt Hub, no joint of its own.
7	Pan Gear, 60 tooth	Carried around by pan; spinning it about the pan axis IS the pan motion.
8	Tilt Gear, 60 tooth	Carried around by pan; spinning it about the tilt axis IS the tilt motion.
9	Pinion Gear, 20 tooth	One copy spins in place to drive pan; the other is carried by pan and spins in place to drive tilt.
10	28BYJ-48 Stepper Motor	One copy is bolted to the fixed plate (never moves); the other is carried by pan (bolted to the tilt bracket).

11	608ZZ Ball Bearing (pan shaft)	Outer ring fixed in the housing; inner ring spins with the pan shaft.
12	625ZZ Ball Bearing (tilt shaft)	Outer ring fixed in the tower (carried by pan); inner ring spins with the tilt shaft.
13	Pan Shaft	Spins about the pan axis -- this rotation IS the pan motion.
14	Tilt Shaft	Carried around by pan; spins about the tilt axis -- this rotation IS the tilt motion.
15	Pan Shaft Spacer	Carried around by pan; spins with the pan shaft.
16	Tilt Shaft Spacer	Carried around by pan; spins with the tilt shaft.

Chapter 3 - How the Motion Is Actually Produced

Both axes use exactly the same trick, just aimed in different directions. A small 20-tooth gear (the pinion, Part 9) is fixed to each motor's own output shaft, and it meshes with a much bigger 60-tooth gear (Part 7 for pan, Part 8 for tilt) fixed to the shaft it drives. A gear is a toothed wheel that locks its teeth into a matching wheel so the two always turn together at a fixed ratio. This is called a **gear reduction**:



Because the big gear has three times as many teeth as the small one, the small gear must spin all the way round three times to make the big gear turn once. That trade means the gear pair also multiplies turning force (**torque** -- a twisting force, like the effort needed to turn a stiff doorknob) by 3x: whatever torque the small gear applies, the big gear delivers three times as much, at one third the speed. Both stages use the same 20-tooth/60-tooth pair, so both axes get an identical 3:1 speed-for-torque trade.

The gear teeth themselves are not modelled with real involute tooth profiles in this file -- each gear is shown as a plain disc sized to the correct pitch diameter for meshing and clearance purposes, which is enough to verify the two gears sit at the right centre distance (measured here at 40.0mm on both stages) and don't collide with anything around them. A manufacturing drawing would need real cut teeth; this design-intent model does not.

Engineering note - the third rotation axis question, now resolved for this design: The original reference file's "Cam Holder" part had pivot bores that lined up with neither the pan axis nor the gear-driven tilt axis -- a genuine open question about whether the original design intended a free-spinning third joint there, that could not be settled from the geometry alone. For Option D, the project owner made an explicit decision: the Payload Fork (Part 6, this design's equivalent part) is built **rigidly bolted** to the Tilt Hub beneath it, with no working joint of its own. It moves only because the Tilt Hub moves. This is a deliberate design choice for this redesign, not a re-discovery of what the original file's geometry actually meant.

A note on overall size: this model measures 109 x 137 x 109 mm (W x H x D), somewhat larger than a literal 1/3 scale of the original (~82 x 90 x 99mm) would give. That overshoot is not a mistake -- it comes directly from the real 28BYJ-48 motor's own fixed can size, mounting-tab spacing, and off-center shaft position, none of which shrink just because the rest of the design does, and this design uses two of them (one per axis). The project owner explicitly accepted this trade-off during planning: the real motor's needs take priority over hitting an exact 1/3 figure.

Chapter 3 (continued) - Assembly

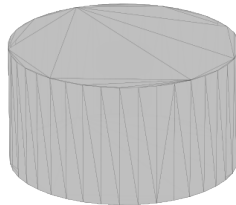
Build the mechanism outward from the housing. This order is inferred from how the parts fit together in the geometry (which surfaces are pressed together vs. simply touching), not from any assembly instructions in the file, since STEP files don't store build order. Two kinds of joints come up throughout:

Fixed / rigid joint - bolted, keyed, or press-fit so the two parts never move relative to each other again.

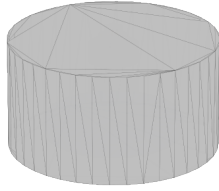
Moving joint - a bearing or a shaft spinning inside a hole, deliberately free to rotate.

Step 1 - Fit the bearings

Press the two 608ZZ bearings into the bottom of the housing (Part 1), and the two 625ZZ bearings into the platform's twin towers (Part 12). All four are moving joints -- their outer rings are fixed in place; their inner rings are what everything else will spin inside.

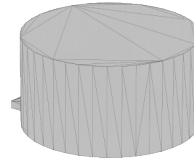


Step 2 - Fit the pan shaft, spacers, and pan gear



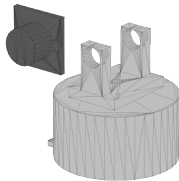
Slide the Pan Shaft (Part 13) up through the two 608ZZ bearings from underneath -- moving joint, it should spin freely once through. Fit the two Pan Shaft Spacers (Part 15) to hold everything at the right height, and lock the Pan Gear (Part 7) onto the shaft's flat-sided section -- fixed joint, the gear and shaft now always turn together.

Step 3 - Add the pan motor



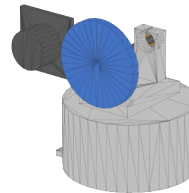
Fix the Pan Motor Mounting Plate (Part 2) in place -- fixed joint, it never moves. Fit the motor body, output shaft, and pinion gear (Parts 9/10, first copies) to the plate and check the pinion meshes cleanly with the pan gear below it.

Step 4 - Add the pan platform and tilt-motor bracket



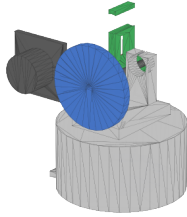
Fit the Pan Platform (Part 3) onto the top of the pan shaft -- fixed joint, it now travels around with pan. Fix the Tilt Motor Mounting Plate (Part 4) to the platform, and fit the second motor/shaft/pinion copy (Parts 9/10) to it.

Step 5 - Fit the tilt shaft, spacers, and tilt gear



Slide the Tilt Shaft (Part 14) through the platform's two 625ZZ bearings -- moving joint. Fit the two Tilt Shaft Spacers (Part 16), then lock the Tilt Gear (Part 8) onto the shaft's flat-sided section and check it meshes cleanly with the tilt pinion.

Step 6 - Fit the Tilt Hub and Payload Fork



Clamp the Tilt Hub (Part 5) onto the tilt shaft's flat-sided section with its grub screw -- fixed joint. Bolt the Payload Fork (Part 6) onto the Tilt Hub's two flanges -- also a fixed joint, by design decision (see Chapter 3): the fork does not have a free-spinning joint of its own, it only moves because the Tilt Hub beneath it moves.

This is a beginner-oriented, plain-language explanation of the Option D pan/tilt CAD model (OPTION_D.step), built as a ~1/3-scale redesign of ASSEMBLY_pan_tilt.STEP around a real, off-the-shelf 28BYJ-48 stepper motor. Every measurement above was taken directly from that file's own 3D geometry, not guessed. This is a design-intent / layout-study model (bought components and gears are simplified stand-ins at the correct size, not release-ready manufacturing geometry) -- real-world fits, tolerances, and print settings should be checked against the STEP file before manufacturing anything from it.